



ETH zürich



**AGENTIC
SYSTEMS** LAB

AGENTIC SYSTEMS LAB

**We design, develop, and evaluate AI systems that run the real world,
and shape the researchers and founders who will define its future.**

Based at ETH Zurich.

Our Team - Researchers, engineers, and builders.

Our team went to...



Our Team - Researchers, engineers, and builders.



Dr. Robert Jakob
Founder & Co-Director
POSTDOC ETH



Dr. Kevin O'Sullivan
Founder & Co-Director
POSTDOC ETH



Dr. Markus Kreft
AI Engineering Lead
POSTDOC ETH



Dr. Patrick Langer
AI Research Lead
AI RESEARCH SCIENTIST



Dr. Fan Wu
Multimodal AI in Healthcare
POSTDOC ETH



Dr. Kevin Riehl
Multimodal AI
PHD RESEARCHER ETH



Pascal Bertrand
Founder, AI Startups and
Collaborations
RESEARCH ASSISTANT



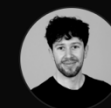
Atoof Shakir
AI Startups and Collaborations
RESEARCH ASSISTANT



Ning Wang
Multimodal AI & RL
PHD RESEARCHER ETH



Thomas Kaar
Multimodal AI
AI RESEARCHER STANFORD



Max Rosenblatti
Multimodal AI
AI RESEARCHER STANFORD



Prof. Dr. Elgar Fleisch
Professor, ETH Zurich & University
of St. Gallen



Dr. Robin Deuber
AI in HCI & Robotics
POSTDOC ETH



David Schaurecker
AI Forecasting
PHD RESEARCHER ETH



Akshaye Sheno
Agentic AI in Healthcare
PHD RESEARCHER NUS



Maximilian May
Agentic AI in Education
PHD RESEARCHER HSG



Vincent Paffrath
AI Transformation
PHD RESEARCHER HSG



Felix Wortmann
Professor & Scientific Director,
University of St. Gallen

Our First Student Batch – The Next Gen of AI Leaders



Gabor Hollbeck
Agentic AI in Science
MASTER STUDENT



Nicolas Zumarraga
Multimodal AI
MASTER STUDENT



Riccardo Mansutti
Agentic AI in Energy
MASTER STUDENT



Baran Peters
AI Evaluation Frameworks
BACHELOR STUDENT



Connor Larson
Agentic AI in Insurance
MASTER STUDENT



Behsad Riemer
AI Infrastructure
BACHELOR STUDENT



Lasse Bærland Strand
Next-Gen RAG & Transformers
MASTER STUDENT



Lorenzo Steno
Agentic AI in Science
MASTER STUDENT



Alpay Hasanli
Agentic AI in Science
MASTER STUDENT



Kilian Bänziger
Next-Gen RAG
MASTER STUDENT



Kacper Ozieblowski
Multimodal AI
MASTER STUDENT



Anirudhh Ramesh
World Models
MASTER STUDENT



Haseeb Raza
AI Research
MASTER STUDENT



Maximilian Weigl
Industry Applications of TSLMs
MASTER STUDENT



Michael Ameri
Multimodal AI in
Telecommunications
MASTER STUDENT



Kalil Sama Bouzigues
Browser Agents
MASTER STUDENT



Francesco Cavalli
Agentic AI for Regulatory Affairs
MASTER STUDENT



Àlex Martí Guix
Multimodal AI
MASTER STUDENT



Joseph Xavier Mertz Echauri
Agentic AI in Finance
MASTER STUDENT



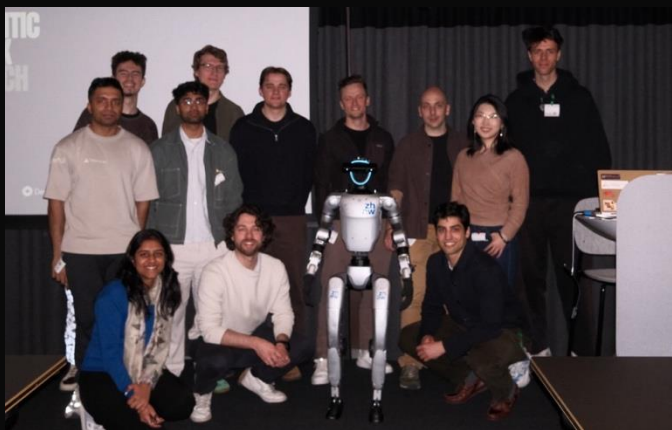
Jakob Flunger
Agentic AI in Insurance
MASTER STUDENT



Marie-Louise Dugua
Recommender Systems
MASTER STUDENT

Our Team – A High-Agency Peer Group That Grows Together

We trust our members to lead, take ownership, and help each other improve.



Join Our Next Student Batch



FOR STUDENTS

AI Research

A predoctoral program designed to turn your thesis into a top-tier CS conference publication with you as first author.

FOR STUDENTS

AI Transformation

Work alongside our startup and industry partners to bring AI applications into leading companies.

FOR STUDENTS

AI Founder

Already working on your own AI idea? Turn your thesis or semester project into a launchpad for your venture.

FOR RESEARCHERS

Affiliate Researcher

For PhD, postdoc, and industry researchers: Pursue collaborative AI research projects with autonomy and mentorship.

We Develop And Evaluate AI systems in Real-World Environments

Our projects span industry verticals and fundamental research, always grounded in practical impact.

01

Agentic AI

We build and evaluate agentic AI applications across a variety of industries, from insurance and finance to education and scientific research.

02

Multimodal AI

We develop Time-Series Language Models (TSLMs) and Audio-Language Models (ALMs) to reason over and predict temporal data in healthcare, manufacturing, telecommunications, and energy.

03

Next-Gen RAG

Next-Gen RAG systems that autonomously plan and optimize retrieval. AutoRAG configures pipelines automatically; MetaRAG orchestrates multiple strategies for reliable, grounded outputs.

04

World Models

We research world models that learn compressed representations of environments, enabling AI systems to simulate, plan, and reason about complex real-world dynamics.

05

AI Evaluation Frameworks

Creating benchmarks and methodologies to measure performance, safety, and reliability of AI Agents and multimodal LLMs.

06

AI Transformation

We research how AI can transform existing enterprise operations. Our work spans user discovery and problem validation to ensure meaningful applications of AI, as well as the design of aligned and controllable AI systems that integrate reliably into real-world workflows.

Introducing Time-Series Language Models (TSLMs)

After sound and vision-language models, our group pioneered the next modality frontier: Time-Series Language Models.

OpenTSLM: Time-Series Language Models for Reasoning over Multivariate Medical Text- and Time-Series Data

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Robert Jakob³ Ning Wang³ Juncheng Liu^{8,9} Aradhana Verma¹⁰ Brian Han¹¹ Daniel Seung Kim¹²
Henry Chubb¹¹ Scott Ceresnak¹¹ Aydin Zahedivash¹ Alexander Tarlochan Singh Sandhu¹⁰
Fatima Rodriguez¹⁰ Daniel McDuff^{1,13} Elgar Fleisch^{2,14} Oliver Aalami¹ Filipa Barata^{2,*}
Paul Schmedmayer^{1,*}

Abstract

Large language models (LLMs) have shown strong capability in interpreting multimodal data but remain limited in their ability to natively handle time-series data. Addressing this limitation could enable the translation of longitudinal and wearable sensing data into actionable insights and patient-facing digital health applications. We propose OpenTSLM, a family of Time-Series Language Models (TSLMs) that integrate time-series as a native modality into pre-trained LLMs, enabling natural-language prompting and reasoning over multiple time-series. We implement two OpenTSLM variants based on soft prompting (OpenTSLM-SoftPrompt) and cross-attention (OpenTSLM-Flamingo). To conduct comprehensive experiments on reasoning over medical text and time-series, we introduce three chain of thought (CoT) datasets: HAR-CoT (human activity recognition), Sleep-CoT (sleep staging), and ECG-QA-CoT (electrocardiogram question answering). Across tasks, OpenTSLM models consistently outperform baselines. OpenTSLMs with time-series encoders trained from scratch achieve 69.88% in sleep staging and 65.44% in HAR, while OpenTSLM combined with time-series foundation models

(TSFMs) achieve 68.33% and 67.64%, compared to 9.05% and 60.44% for fine-tuned text-only baselines. Additionally, we conduct expert evaluations with cardiologists, which show that OpenTSLMs exhibit strong reasoning capabilities and temporal understanding on raw sensor data for ECG-QA. We further show that OpenTSLM-Flamingo models scale better in memory as the number and length of time-series increase. To facilitate further research, we release all code, datasets, and models as open-source resources.

1. Introduction

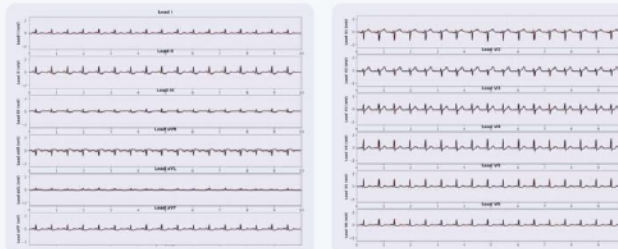
Medicine is inherently temporal: assessment, diagnosis, and treatment depend on how signs, symptoms, and biomarkers evolve over time (Giannoula et al., 2018; Henly et al., 2011; Jørgensen et al., 2024). Clinical decision-making relies on temporal patterns—tracking vital signs, medication responses, laboratory values, and disease progression interventions to guide diagnosis, prognosis, and therapeutic interventions. As time-series data from electronic health records and continuous monitoring proliferate (Abernethy et al., 2022; Marra et al., 2024; Yeung et al., 2023), human-legible representations become essential for interpreting and managing this information (Olex & McInnes, 2021; Senathirajah et al., 2020; Zhou et al., 2008). Clinical summaries must translate complex temporal patterns, e.g., biomarker trajectories, into interpretable assessments that support evidence-based decision-making and care coordination.

Recent advances in multimodal large language models (LLMs) allow users to interpret complex data through natural language, synthesizing information across text, images, audio, and video (Wu et al., 2023; AlSaad et al., 2024). However, reasoning over time-series data remains a critical blind spot among currently supported modalities. Prior work has attempted to integrate time-series as plain text tokens (Gruver et al., 2023; Kim et al., 2024; Liu et al., 2023); however results have been limited (Merrill et al.,

Proven on 12-lead ECGs with interpretable CoT rationales validated by cardiologists.

Prompt

What range does the p duration of this 12-lead ECG <TS>, ..., <TS> fall in? Answers: 1. above the normal range 2. below the normal range 3. within the normal range



Clinical Context: 74-year-old female patient. 12-lead ECG. clinical recording. recorded with AT-60 3. Signal quality: baseline drift noted, static noise pre-sent, burst noise present, electrode artifacts present. extra beats detected. pacemaker present.

✧ Rationale

To determine the P wave duration, we must first consider the standard normal range, which is typically 0.08 to 0.11 seconds. In this ECG, despite the presence of baseline drift, static noise, burst noise, and electrode artifacts, the P waves can still be evaluated. The presence of a pacemaker may influence the rhythm, but it does not directly affect the intrinsic P wave duration. The P waves appear consistent across the leads, and even with the noted artifacts, they do not seem prolonged. Given the patient's age and the presence of a pacemaker, it is crucial to assess whether any conduction abnormalities could affect the P wave duration. However, the morphology and duration of the P waves do not suggest any significant prolongation. Therefore, considering the observed patterns and clinical context, the P wave duration falls within the normal range. **Answer: within the normal range**

OpenTSLM Paper



OpenTSLM Github



★ Starred 1.2k

*Equal contribution ¹Stanford Mussallem Center for Biodesign, Stanford University, USA ²Centre for Digital Health Interventions, ETH Zurich, Switzerland ³Agentic Systems Lab, ETH Zurich, Switzerland ⁴University of Illinois Urbana-Champaign, USA ⁵Google Research, USA ⁶Stanford University, USA ⁷Amazon, USA ⁸Microsoft ⁹National University of Singapore, Singapore ¹⁰Division of Cardiovascular Medicine, Stanford University, USA ¹¹Pediatric Cardiology, Stanford University, USA ¹²Division of Cardiology, University of Washington, USA ¹³University of Washington, USA ¹⁴Centre for Digital Health Interventions, University of St. Gallen, Switzerland. Correspondence to: Patrick Langer <planger@stanford.edu>.

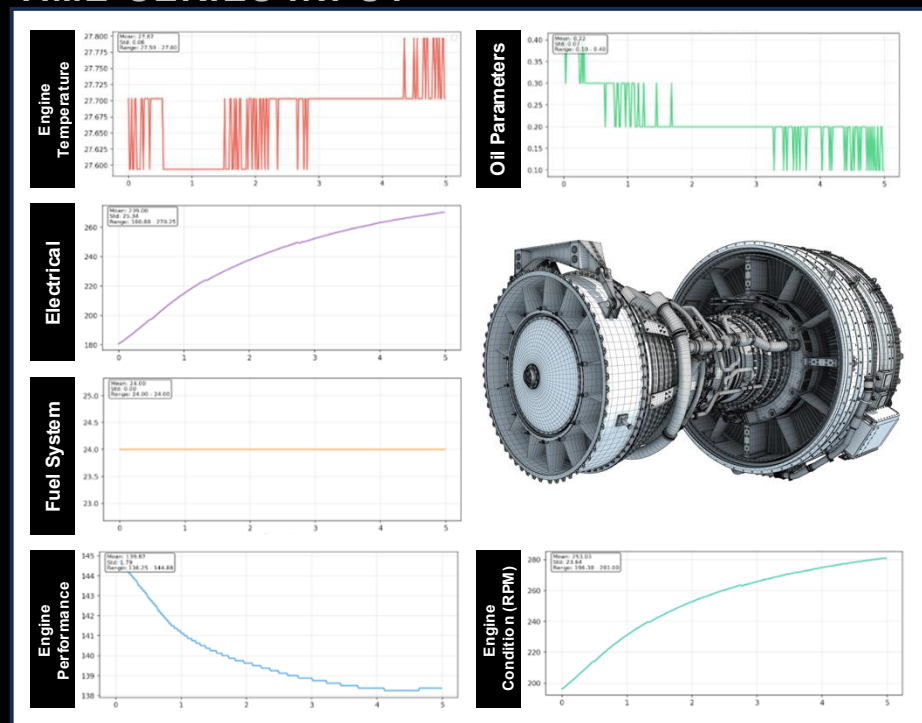
Preprint. February 17, 2026.

Give AI a Sense of Time

PROMPT INPUT

1. Analyze the sensor patterns to identify potential issues or anomalies
2. Recommend specific maintenance actions to address the problems
3. Predict the expected outcome if those maintenance actions are taken

TIME-SERIES INPUT



STATE SUMMARY

During the first 5 minutes, RPM and electrical **rise smoothly**, but CHT shows **irregular step-like oscillations with brief spikes** (~0.15 units) and a slow upward drift not correlated with power or mixture. EGT trends downward despite increasing RPM, and oil system signal decays with jitter. The **combination points to unstable cooling airflow** around a cylinder-consistent with a loose/damaged baffle tie/tie rod causing baffle flutter and leakage.

RECOMMENDED ACTION

Inspect engine cooling baffles, tie/tie-rod hardware, and seals around the affected cylinder(s). Tighten or replace the baffle tie/tie-rod, restore baffle seals, and **secure all fasteners and brackets**.

EXPECTED OUTCOME

CHT will **stabilize with a lower average and reduced variability** (spikes eliminated, <+0.03 units), EGT will track power more normally (no persistent decline), and **oil system readings will stabilize without jitter**.

STEP-BY-STEP REASONING

GPT 5.5 · 1M context



OpenTSLM

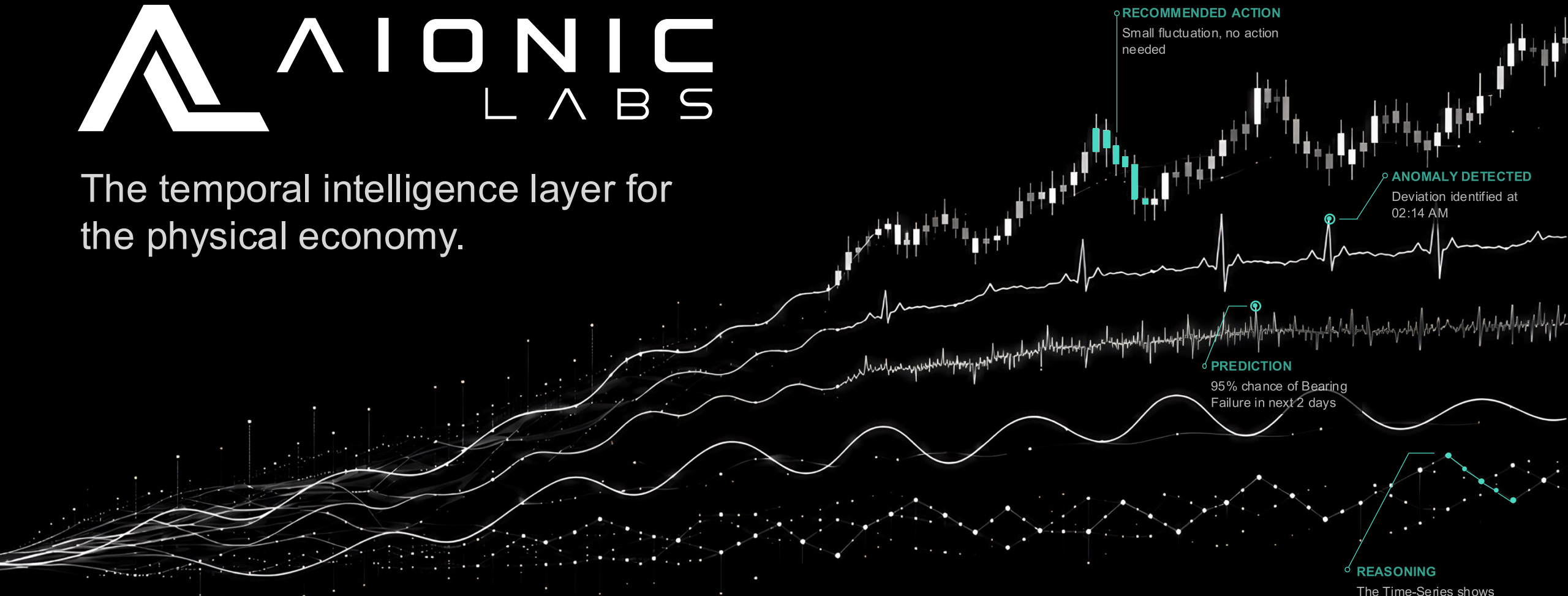
~25 min



At ~200K tokens / 5 min, a text window is spent in ~25 minutes. OpenTSLM streams **long, multivariate signal** natively.



The temporal intelligence layer for
the physical economy.



RECOMMENDED ACTION

Small fluctuation, no action
needed

ANOMALY DETECTED

Deviation identified at
02:14 AM

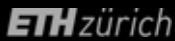
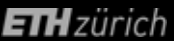
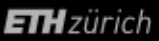
PREDICTION

95% chance of Bearing
Failure in next 2 days

REASONING

The Time-Series shows
unusual degradation outside
of established patterns

Join Our Group. From Europe, for the World.

Dr. Patrick Langer RESEARCH & TRAINING INFRA Meta Superintelligence · Stanford First author of OpenTSLM    	Dr. Martin Maritsch ENGINEERING & OPERATIONS AWS Senior GenAI lead Stanford · ETH Medal    	Dr. Robert Jakob GTM, SALES & FUNDRAISING ETH Agentic Systems Lab Founder & Director · Harvard    	Thomas Kaar PRODUCT & GTM Ex-Meta · Stanford · CDTM OpenTSLM core architect    
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65+ RESEARCH CONSORTIUM	TEAMS BEHIND	MOIRAI Salesforce	Chronos Amazon	GIFT-Eval Salesforce	Large Sensor Model Google	SensorFM Google	TimeCopilot
	ACROSS	         					
	          						



Join our open
research
community

<https://github.com/OpenTSLM>

Applications of TSLMs.



Use cases are wherever people act on signals. Our models learn to reason over time-series the way a human would.

FROM AGENTIC AI TO TEMPORAL AI

Temporal AI Challenge

AI that understands how the world changes over time.

AIONIC
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ETH zürich

 **AGENTIC**
SYSTEMS LAB



Give AI a Sense of Time

Use Aionic's TimeNet to prepare open-source time-series data, then train a TSLM that connects temporal signals with language for a useful task.

- 01 INTERPRET
- 02 ANTICIPATE
- 03 ACT



HEALTH

A patient's condition changes



INDUSTRY

A machine begins to fail



ENERGY

Demand shifts over time

Use cases are wherever people act on signals. Our models learn to reason over time-series the way a human would.

Give AI a Sense of Time

File Account Help

New Window IBot FYIs

Order Entry

ADBE

300

145.75 146.68

Adaptive

Option Chain

Buy Sell

QTY 100 LMT LMT PRICE 0.00 DAY

Submit

ADBE

5 min candles

File Edit View

146.61

+0.25 0.17%

Bid 125 x 145.75

Ask 10 x 146.68

Today High 146.79

Today Low 145.57

52 Wk High 147.45

52 Wk Low 95.42

EARNINGS

EPS 2.88

P/E 51.1

VALUATION

Market Cap 72.5B

Price/Book 9.3

IV/OI

Opt Open Int 242K

10/Q

ADBE

HL

Jul 17 10:00 11:00 12:00 13:00 14:00

Jul 11 Jul 12 Jul 15 Jul 16 Jul 17

Activity Orders Trades Summary

ACCOUNT ACTION TYPE DETAILS QUANTITY FILL PRICE

IBKR DCU00060 Buy STP LMT STP 62.75 0/100 Cancel

JNJ DCU00060 Buy LMT LMT 134.23 0/100 Cancel

GILD DCU00060 Buy LMT LMT 71.05 100 71.05

CRM DCU00057 Sell MKT 396 89.16

AMAT DCU00057 Sell MKT 25 46.22

TXN DCU00054 Buy LMT LMT 81.79 25 81.79

Tech · Big Tech

Which company has the best AI model end of May?

Past May 31 Jun 30

Anthropic 97.3% Google 1.1% OpenAI 1.1% ByteDance <1%

Polymarket

100% 75% 50% 25% 0%

Apr 19 May 3 May 17

\$10,000,174 Vol. May 31, 2026

1H 6H 1D 1W 1M ALL

AI Anthropic \$845,797 Vol. 97% 40% Buy Yes 97.3¢ Buy No 2.8¢

Google \$925,360 Vol. 1% 20% Buy Yes 1.1¢ Buy No 99.0¢

AI Which company has the best AI m... Anthropic · Yes

Buy Sell Market

Yes 97.3¢ No 2.8¢

Amount \$0.00 cash \$0

+\$1 +\$5 +\$10 +\$100

Buy Yes

By trading, you agree to the Terms of Use.

All Tech AI Big Tech Rewards 20

AI Will Anthropic have the second best AI model at the end of May 2026? 97%

AI Will Anthropic have the third best AI 59%

From Problem to Proof

01



PROBLEM + DATA

Identify a target user and **problem**, then source open data. Where useful, build an agentic pipeline to search, assess, select, retrieve, or validate datasets.

Ideal Data Profile

- A dataset with time-series and text reasoning over it / or describing it
- In some cases, you might consider generating text annotations with LLMs (as in OpenTSLM paper)

02



CONNECT

Use or build a TimeNet connector for signals, metadata, and annotations. Define the questions, labels, or other learning targets. **Reusable ingestion pipelines** are especially valuable.



Open infrastructure that standardises time-series data and metadata for TSLM training.

03



TRAIN + EVALUATE

Train or fine-tune a TSLM. Compare it with a baseline on held-out data, avoiding leakage across time, subjects, or devices.

04



DEMONSTRATE

Show how you sourced the data, real inputs, model outputs, and how the result helps the target user. Explain the **evidence and limitations**.

What You Get, What You Submit

YOUR RESOURCES

- TimeNet access
 - Access to training compute
 - OpenTSLM as a reference
 - Technical guidance from Aionic Labs + ASL
- Discord

YOU SUBMIT

- Working demo
- Code + training configuration
- Checkpoint or adapter
- Dataset documentation
- Short evaluation with baseline comparison

FINAL PRESENTATION Live demo to the jury: problem, approach, results, and key learnings

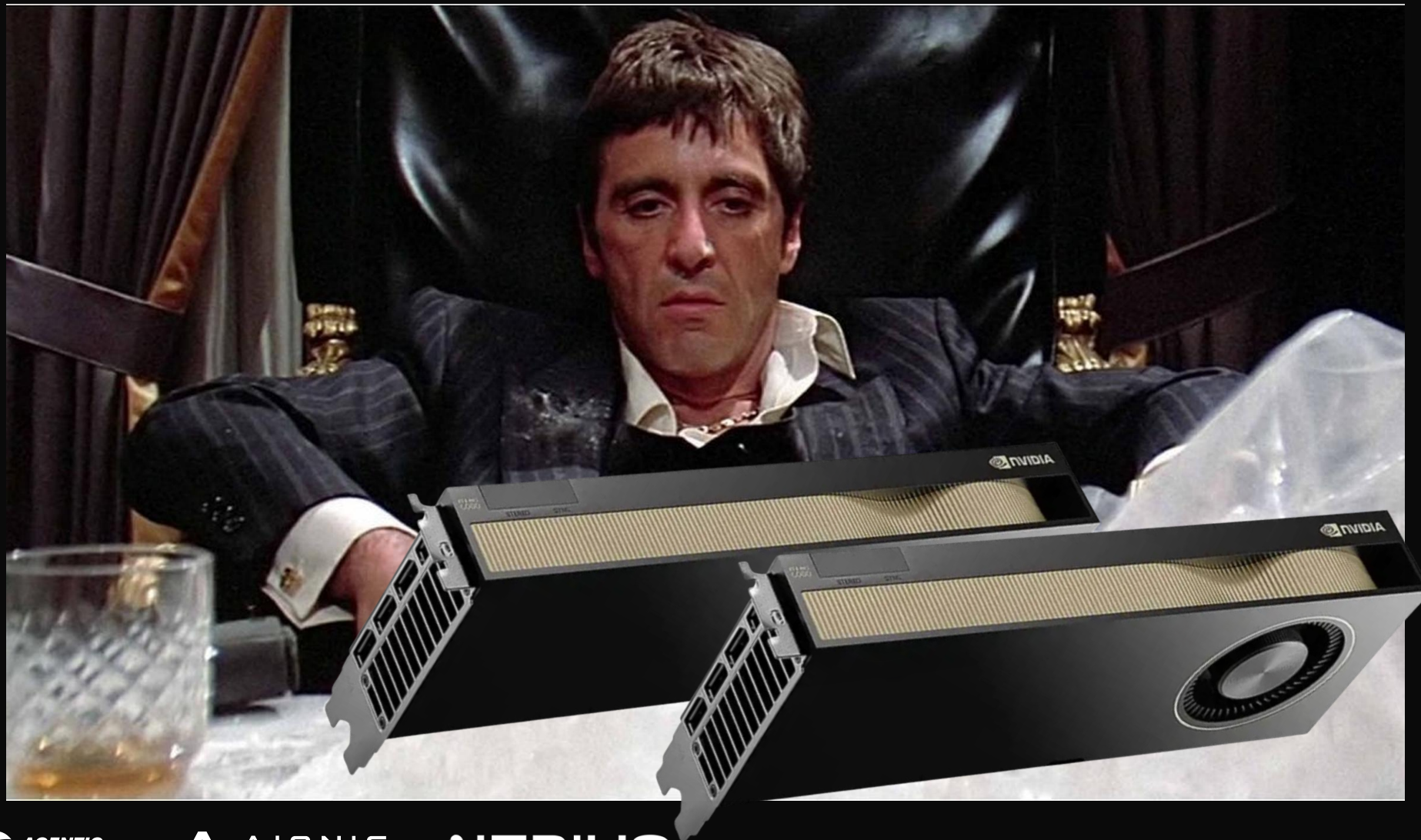
WHAT MAKES A STRONG PROJECT

- A useful problem
- Thoughtful data preparation
- Adequate TSLM training
- Rigorous evaluation
- A clear demonstration

BONUS POINTS

- Well-chosen datasets and reusable TimeNet connectors that others can build on.
- Agentic pipelines for automated search, assessment, selection, retrieval, or validation of datasets are a plus

We need more...



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NEBIUS

\$1000 compute voucher per team
...and more for the winners

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NEBIUS

Nebius: the full-stack AI cloud.

Global AI cloud · HQ Amsterdam · NASDAQ: NBIS · GPUs to inference & agentic search · deep NVIDIA partnership

5 GW

contracted power target

1 GW+

added every year from 2027

Full stack

GPUs, inference & search

Newest GPUs

Blackwell Ultra live today

New in Zurich

The Zurich office opens mid-September
Nebius' agentic search hub with the
Tavily team on board, and home of the
DACH startup crew.

Say hi to Jennifer Werthwein!

Fuel for builders

Startup credits, research grants, the
Builder Program and the Early Talent
Program (winter cohort Jan 2027)

Start at nebius.com/startups.

This weekend

You can train on Nebius GPUs with
sponsored vouchers

The most promising teams get follow-on
credits to keep building after Sunday.

Not just infrastructure. A complete platform for anyone building AI products, agents or services.

Credits, grants and careers - three ways in.

NEBIUS Programs for founding, researching or graduating

Nebius for Startups

NVIDIA Inception members and selected VCs can nominate startups for the credit program.

Including vertical tracks, e.g. the Robotics & Physical AI Awards with compute and inference credits.

- nebius.com/startups

Research & builders

Research grants: compute credits via Nebius Academy. Nebius Fellows for individual builders.

Builder Program (early preview): Token Factory & Tavily credits, training, office hours and community.

- nebius.com/nebius-research-grants
- dev.nebius.com/builders
- nebius.com/fellows

Early Talent Program

Recent grads & final-year students · 2–6 months, office-based, paid like a junior full-timer.

Real product ownership from day one, a dedicated mentor (and maybe a full-time offer.)

Winter cohort Jan 2027 → apply by Dec 31 (Paris, Amsterdam, Tel Aviv, Finland, US)

- nebius.com/early-talent-program

Nebius supports you building with credits, compute and your first job.

Resources

EVERYTHING YOU NEED TO GET STARTED

European Hackathon League ehl.gg

Aionic Labs aioniclabs.ai

Agentic Systems Lab agenticssystemslab.org

TimeNet documentation docs.timenet.ai

TimeNet on GitHub github.com/OpenTSLM/TimeNet

OpenTSLM paper arxiv.org/abs/2510.02410

DISCORD SUPPORT

Questions during the
hackathon?

Scan to join the support channel.



discord.com/invite/qXsAt4BHV